

Real-Time Data Ingestion & Processing using Apache NiFi

1. Introduction

This project demonstrates the design and implementation of a complete real-time data ingestion and processing pipeline using Apache NiFi, PostgreSQL, Docker, and Hadoop HDFS.

The main objective of this assignment was to simulate a real-world streaming environment where data is continuously generated from multiple sources, processed through different transformation stages, and finally stored in a distributed storage system.

The project focuses on applying practical data engineering concepts including:

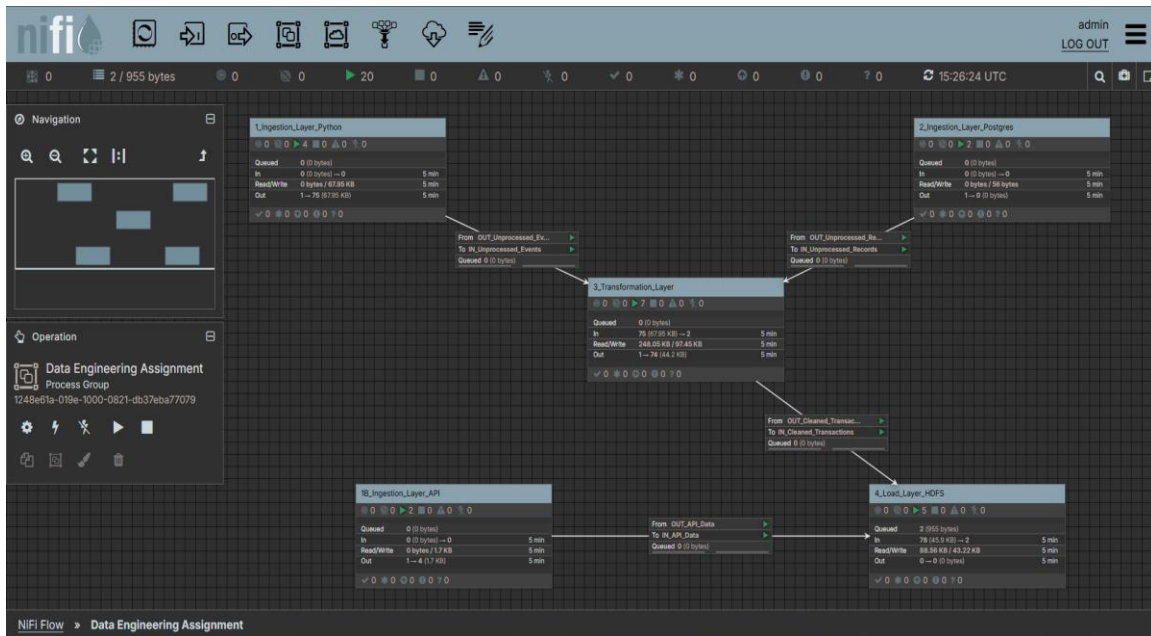
- Real-time ingestion
- Data cleaning
- Deduplication
- Incremental database extraction
- API ingestion
- Distributed storage using HDFS
- Apache NiFi best practices

2. Project Architecture

The pipeline architecture was divided into five logical layers to improve modularity, readability, and scalability.

Architecture Layers:

1. Python File Ingestion Layer
2. API Ingestion Layer
3. PostgreSQL Ingestion Layer
4. Transformation Layer
5. HDFS Load Layer



3. Real-Time Data Simulation

A custom Python script was developed to simulate real-time transaction events.

The script continuously generates JSON files every few seconds and stores them inside a monitored directory.

The generated data intentionally contains several data quality issues to simulate real-world datasets, including:

- Missing IDs
- Invalid amount values
- Duplicate records
- Inconsistent status values

```

1  [
2    {
3      "id": "12642b29-c04c-4f9d-a877-2a452c4d53f4",
4      "user_id": 1229,
5      "amount": 427.71,
6      "status": "Failed",
7      "timestamp": "2026-05-10T20:01:00.686421",
8      "category": "Books"
9    },
10   {
11     "id": "b7781714-a567-496d-bb8d-4668623d5058",
12     "user_id": 1524,
13     "amount": 1044.4,
14     "status": "???",
15     "timestamp": "2026-05-10T20:01:00.686421",
16     "category": "Books"
17   },
18   {
19     "id": "c83974bf-f9f4-49bb-a100-9d929dcf704",
20     "user_id": 4609,
21     "amount": 631.06,
22     "status": "Pending",
23     "timestamp": "2026-05-10T20:01:00.686421",
24     "category": "Tech"
25   }
26 ]

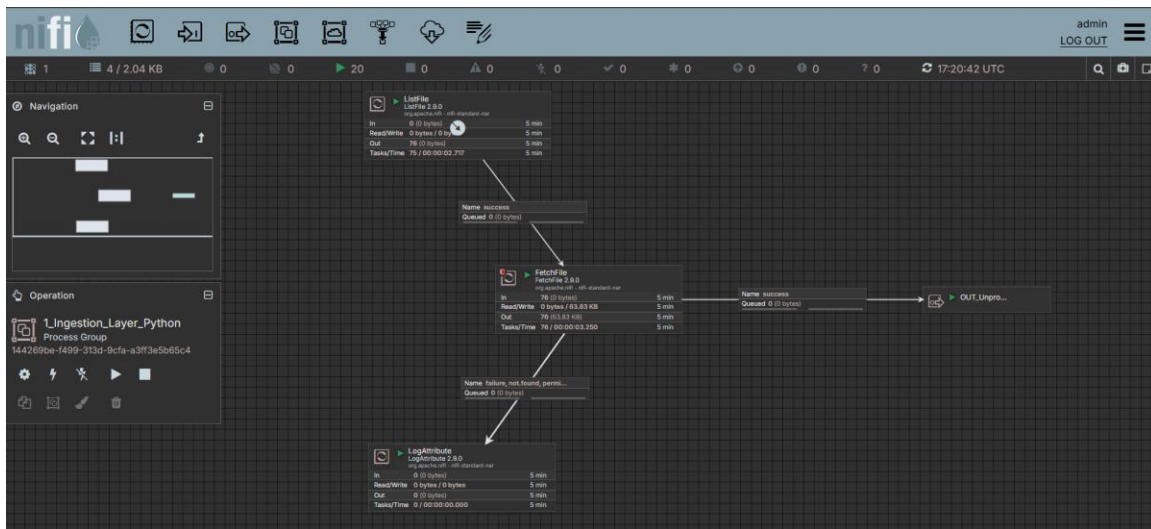
```

4. File-Based Ingestion using Apache NiFi

The mandatory ingestion pattern required in the assignment was implemented successfully using:

ListFile → FetchFile

The ListFile processor monitors the input directory continuously, while FetchFile retrieves the actual file content and deletes processed files to prevent duplicates.



5. PostgreSQL Database Ingestion

A PostgreSQL container was integrated into the environment using Docker Compose.

A sample table named 'db_transactions' was created to simulate structured enterprise data.

NiFi connected to PostgreSQL using DBCPConnectionPool and QueryDatabaseTableRecord with incremental extraction enabled.

The screenshot shows the pgAdmin 4 interface. On the left, the 'Servers' tree is expanded to show 'nifi_database/nifi_user@NiFiAssignment_DB'. The main pane displays a SQL query in the 'Query' tab:

```

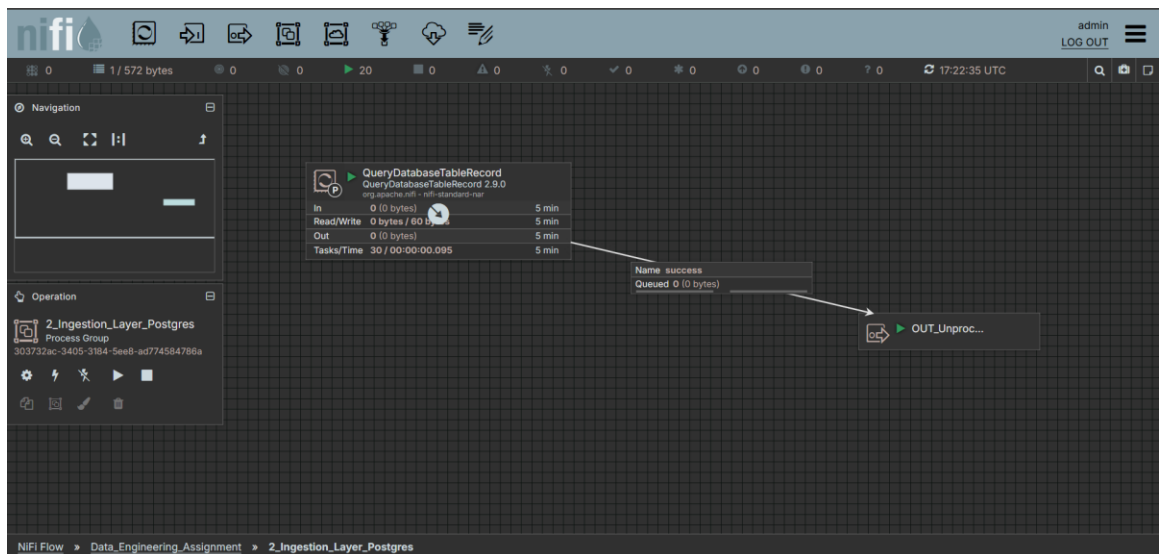
1 CREATE TABLE db_transactions (
2   id SERIAL PRIMARY KEY,
3   user_id INT,
4   amount VARCHAR(50),
5   status VARCHAR(50),
6   category VARCHAR(50),
7   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
8 );
9
10 INSERT INTO db_transactions (user_id, amount, status, category) VALUES
11 (101, '250.50', 'Active', 'Electronics'),
12 (102, 'INVALID', '???', 'Books'),
13 (103, '100.00', 'Pending', 'Clothing'),
14 (104, NULL, 'Failed', 'Food'),
15 (105, '500.75', 'Active', 'Electronics');
16
17
18 Select * from db_transactions;

```

Below the query, the 'Data Output' tab shows the results of the query in a table. The table has 8 columns: id, user_id, amount, status, category, and created_at. The data is as follows:

id	user_id	amount	status	category	created_at
1	1	101	250.50	Active	Electronics
2	2	102	INVALID	???	Books
3	3	103	100.00	Pending	Clothing
4	4	104	[null]	Failed	Food
5	5	105	500.75	Active	Electronics

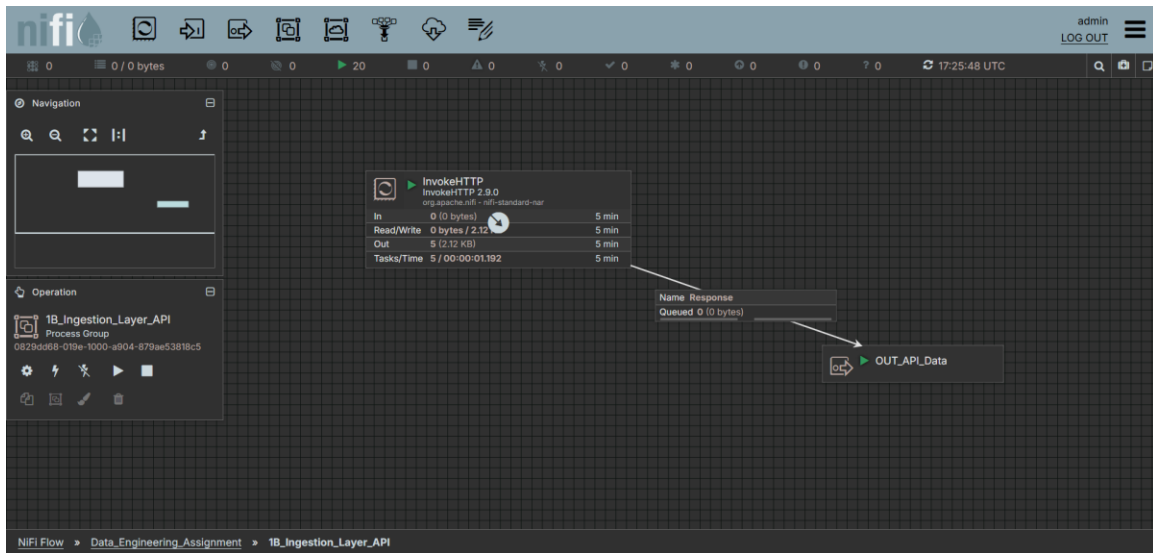
The bottom status bar indicates 'Total rows: 5' and 'Query complete 00:00:00.654'.



6. API Ingestion Layer

The pipeline retrieves real-time market exchange data using the InvokeHTTP processor from the Frankfurter Exchange Rate API.

This demonstrates how Apache NiFi can integrate external APIs into streaming pipelines.



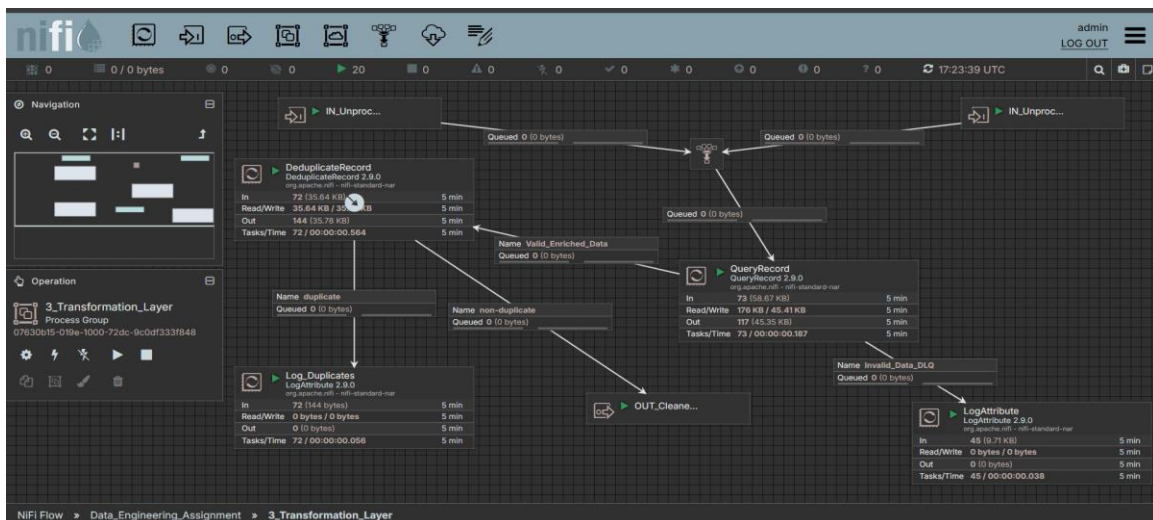
7. Data Transformation Layer

The transformation layer was responsible for improving data quality before storage.

Transformations included:

- Data cleaning
- Data validation
- Deduplication
- Data type conversion
- Data enrichment

The DeduplicateRecord processor was used with SHA-256 hashing to remove duplicate transactions.



8. Loading Data into HDFS

The final processed datasets were stored inside Hadoop Distributed File System (HDFS).

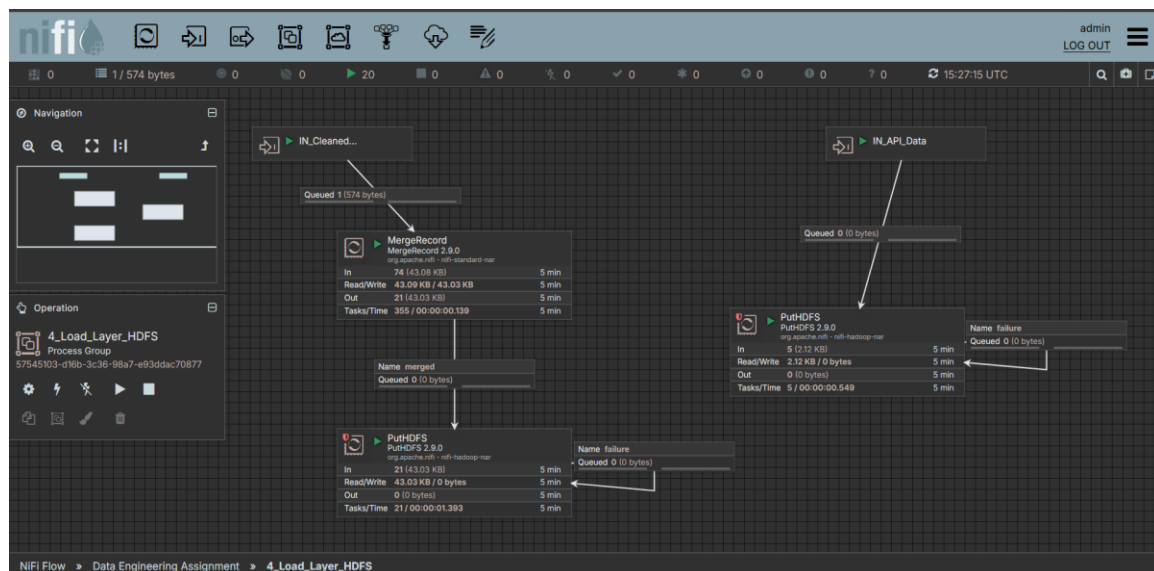
The following processors were used:

- MergeRecord
- PutHDFS

Data was stored inside:

/user/itversity/transactions

/user/itversity/market_data



9. NiFi Best Practices Applied

The project applied several Apache NiFi best practices, including:

- Back Pressure
- Retry and Penalization
- Relationship handling
- Incremental processing
- Modular Process Groups

Incremental Processing (State Management): To optimize resource utilization and achieve true real-time streaming, the pipeline relies heavily on incremental processing.

- **Database Ingestion:** The PostgreSQL ingestion processor utilizes **Maximum-value Columns** (State Management) to keep track of the last retrieved row.

Instead of executing full table scans, it appends only newly inserted or updated records, significantly reducing database load.

- **File Ingestion:** The `ListFile` processor maintains the internal state of directory listings, ensuring that previously ingested JSON files generated by the Python script are not duplicated, pulling only fresh data as it arrives.
- **Modular Process Groups (Separation of Concerns):** To avoid the common anti-pattern of a cluttered "spaghetti flow," the pipeline was architected using Modular Process Groups.
- The entire ETL lifecycle is logically separated into distinct layers: Ingestion (handling Python scripts, APIs, and PostgreSQL independently), Transformation (centralized cleaning and routing), and Load (HDFS integration). This modular design pattern ensures high readability, simplifies debugging, and allows each layer to be scaled or modified independently without impacting the rest of the pipeline

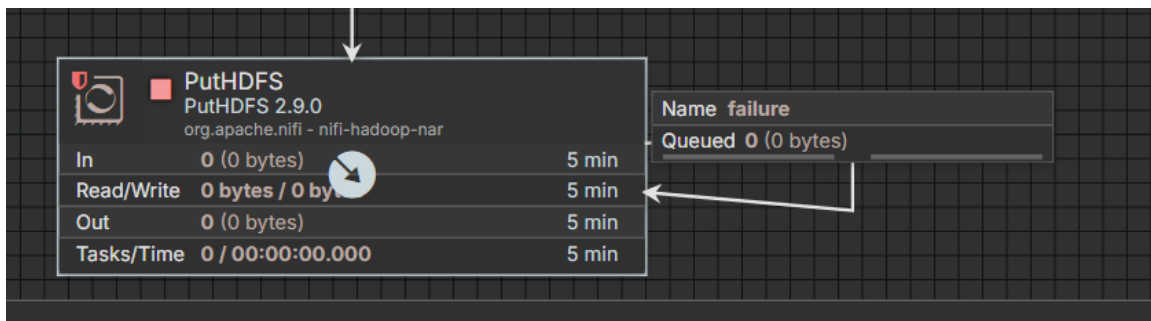
Back Pressure Configuration: The default object threshold was explicitly reduced from 10,000 to 5,000. This was a deliberate architectural decision to prevent Out-Of-Memory (OOM) errors and ensure container stability, considering the heavy concurrent resource utilization by Hadoop and PostgreSQL within the local Docker environment.

The screenshot shows the 'Edit Connection' dialog box with the 'Settings' tab selected. The 'Details' tab is also visible. The 'Settings' tab includes sections for 'Available Prioritizers' and 'Selected Prioritizers'. The 'Details' section contains fields for Name, Id, FlowFile Expiration, Back Pressure Object Threshold, Size Threshold, and Load Balance Strategy.

Details	Settings
Name	Available Prioritizers ⓘ
Id cadd3443-eac2-3ff1-4639-d09b9578528c	FirstInFirstOutPrioritizer
FlowFile Expiration ⓘ* 0 sec	NewestFlowFileFirstPri...
Back Pressure Object Threshold ⓘ* 5000	OldestFlowFileFirstPrio...
Size Threshold ⓘ* 1 GB	PriorityAttributePrioriti...
Load Balance Strategy ⓘ* Do not load balance	Selected Prioritizers ⓘ

Cancel Apply

Penalization Logic: To ensure data reliability, a **Retry Loop** was implemented on the `PutHDFS` processor by routing the `failure` relationship back to the processor itself. This allows the system to handle transient network issues or destination downtime. To prevent system resource exhaustion during failures, a **Penalization** of 30 seconds was applied (Penalty Duration). This ensures that failed FlowFiles are "penalized" and held back before the next retry, giving the destination system time to recover.



Edit Processor | PutHDFS 2.9.0

Settings

Scheduling

Properties

Name*

PutHDFS

Id

02ae9699-74fa-31dc-1b8c-f6d40ad9514e

Type

PutHDFS 2.9.0

Bundle

org.apache.nifi - nifi-hadoop-nar

Penalty Duration ⓘ*

30 sec

Yield Duration ⓘ*

1 sec

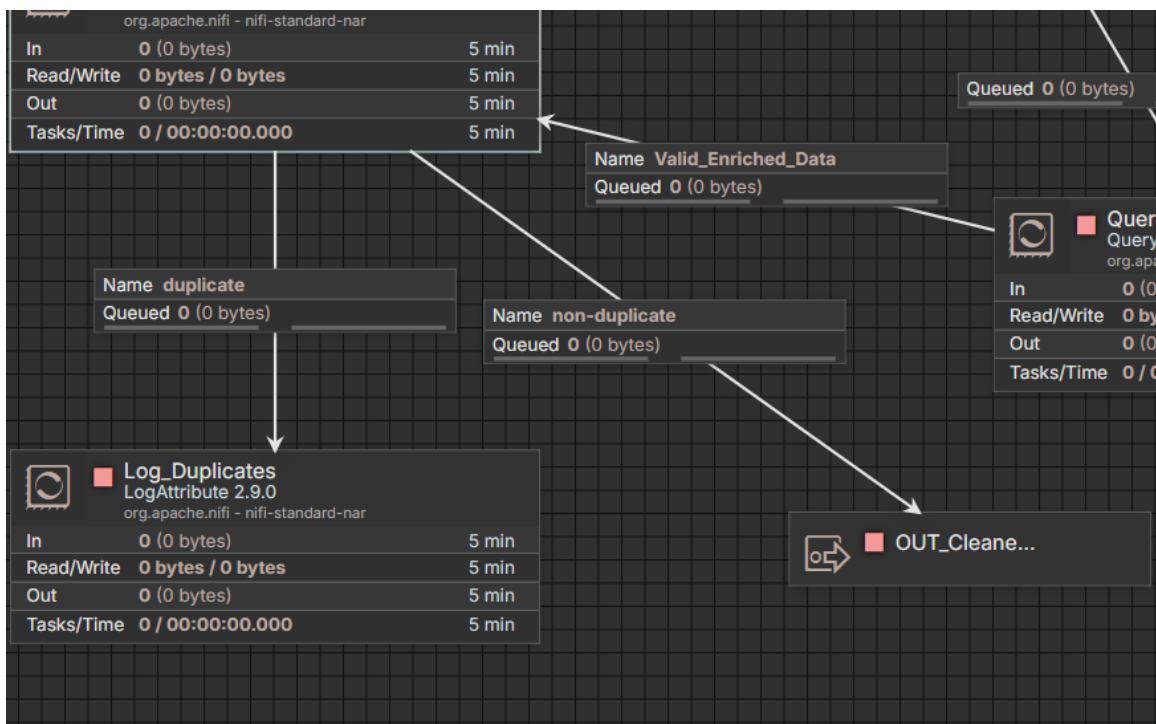
Bulletin Level ⓘ*

WARN

Stopped ▼

Comprehensive Relationship Handling: The pipeline avoids "dead-ends" by explicitly defining the path for every possible processing outcome.

- **In the Transformation Layer:** Using `DeduplicateRecord`, data is bifurcated into `non-duplicate` for further processing and `duplicate` for logging and termination.
- **In the Validation Stage:** Invalid records are routed to a separate path to ensure downstream data quality.
- **Error Handling:** All `failure` relationships are either handled via retry loops or recorded, ensuring zero data loss and maintaining the integrity of the data stream.



10. Challenges Faced and Solutions

Challenge 1 — Missing PutHDFS Processor

The latest Apache NiFi version did not include the PutHDFS processor by default.

Solution:

Custom Hadoop NAR extensions were manually added to NiFi.

Challenge 2 — HDFS Connection Refused Error

The original Hadoop configuration used localhost which prevented communication between Docker containers.

Solution:

The configuration was changed from:

`hdfs://localhost:9000`

to:

hdfs://itvdelab:9000

Challenge 3 — Missing HDFS Directories

The required HDFS directories did not exist initially.

Solution:

The directories were created manually using Hadoop DFS commands.

11. Results

The final pipeline successfully achieved:

- Real-time ingestion
- Multi-source ingestion
- Data transformation
- Deduplication
- Incremental extraction
- API integration
- HDFS storage

```
PS C:\Users\moham\Desktop\week 8> docker exec -it itvdelab_assignment bash
itversity@itvdelab:~$ hdfs dfs -ls /user/itversity/transactions
Found 39 items
-rw-r--r-- 3 nifi supergroup 1894 2026-05-10 15:23 /user/itversity/transactions/06f3c806-1601-4737-9bc0-05a78fdbee44
-rw-r--r-- 3 nifi supergroup 2269 2026-05-10 15:21 /user/itversity/transactions/0b2b9170-1717-4fb8-aa3b-61620f75832f
-rw-r--r-- 3 nifi supergroup 2088 2026-05-10 15:21 /user/itversity/transactions/0f4eec51-984a-42e3-9de5-7e85adb0d779
-rw-r--r-- 3 nifi supergroup 2088 2026-05-10 15:21 /user/itversity/transactions/1200e805-1b2a-4423-a889-8cd6f5d24547
-rw-r--r-- 3 nifi supergroup 2460 2026-05-10 15:21 /user/itversity/transactions/16832a8f-7724-4e08-a53f-aec4531c6ca0
-rw-r--r-- 3 nifi supergroup 2657 2026-05-10 15:21 /user/itversity/transactions/169736c1-0e35-44c2-a0bd-77ef4428a091
-rw-r--r-- 3 nifi supergroup 2279 2026-05-10 15:22 /user/itversity/transactions/186ca0ac-ed48-49c9-b05d-53973fa34451
-rw-r--r-- 3 nifi supergroup 2466 2026-05-10 15:23 /user/itversity/transactions/1d43c37c-850f-423f-bf2f-e1bc3abfbce7
-rw-r--r-- 3 nifi supergroup 34305 2026-05-10 15:21 /user/itversity/transactions/1f5c9f9d-8296-4a9b-a66a-a820327f8e04
-rw-r--r-- 3 nifi supergroup 2098 2026-05-10 15:21 /user/itversity/transactions/22f59a1a-f8eb-4ae7-b345-ae01a4ff2516
-rw-r--r-- 3 nifi supergroup 38890 2026-05-10 15:21 /user/itversity/transactions/2bd0d9b4-f708-44ae-b5c6-3f709b3a71bb
-rw-r--r-- 3 nifi supergroup 1891 2026-05-10 15:22 /user/itversity/transactions/2d61ae41-a06d-48cb-9002-8e3bde4a4e4f
-rw-r--r-- 3 nifi supergroup 2280 2026-05-10 15:21 /user/itversity/transactions/4033b720-677e-4a11-9212-60fecd7ab065
-rw-r--r-- 3 nifi supergroup 2273 2026-05-10 15:21 /user/itversity/transactions/48faefaa-d67c-44d1-862a-922fc428e7d2
-rw-r--r-- 3 nifi supergroup 2472 2026-05-10 15:21 /user/itversity/transactions/550d551b-2662-48b5-b10b-ea36a740a824
-rw-r--r-- 3 nifi supergroup 1907 2026-05-10 15:21 /user/itversity/transactions/60b6465e-fd21-4f0e-b91e-f7cc3b8ac24a
-rw-r--r-- 3 nifi supergroup 2099 2026-05-10 15:21 /user/itversity/transactions/65f31f50-ce50-4dfc-9f5a-4b620e6befcb
-rw-r--r-- 3 nifi supergroup 2851 2026-05-10 15:21 /user/itversity/transactions/6add4c0f-e305-4e1e-a9cc-f300e0442f1d
-rw-r--r-- 3 nifi supergroup 3043 2026-05-10 15:21 /user/itversity/transactions/6af669a4-4b40-4852-8728-dcf4fce625fd
-rw-r--r-- 3 nifi supergroup 2084 2026-05-10 15:21 /user/itversity/transactions/6ca02827-f4db-4635-ae9e-4e0340cbe092
-rw-r--r-- 3 nifi supergroup 2464 2026-05-10 15:21 /user/itversity/transactions/732fb6de-9cff-43bf-97b5-86e36ddfddefc
-rw-r--r-- 3 nifi supergroup 2471 2026-05-10 15:21 /user/itversity/transactions/779671d3-9e44-4b26-933c-b3122233c4f4
-rw-r--r-- 3 nifi supergroup 1705 2026-05-10 15:21 /user/itversity/transactions/7804facf-4a78-45c8-8372-276d25bc59d7
-rw-r--r-- 3 nifi supergroup 1714 2026-05-10 15:23 /user/itversity/transactions/7e168ffb-f453-45b7-8d71-298857873dd9
-rw-r--r-- 3 nifi supergroup 1896 2026-05-10 15:21 /user/itversity/transactions/8a035867-8bb3-4c64-930a-1643fac2bbe7
-rw-r--r-- 3 nifi supergroup 1911 2026-05-10 15:23 /user/itversity/transactions/910681ef-373d-44a6-8fb3-aa8e5d09a7db
-rw-r--r-- 3 nifi supergroup 2454 2026-05-10 15:21 /user/itversity/transactions/9bbe5cc0-7947-4289-a700-66cf7345a2bc
-rw-r--r-- 3 nifi supergroup 945 2026-05-10 15:21 /user/itversity/transactions/Transaction_20260510_175454.json
-rw-r--r-- 3 nifi supergroup 1145 2026-05-10 15:21 /user/itversity/transactions/a769e5b7-015b-46fb-835d-deb70e7f0c36
-rw-r--r-- 3 nifi supergroup 22967 2026-05-10 15:21 /user/itversity/transactions/a971a92a-8824-4abf-bea3-9dbe52de2cf9
-rw-r--r-- 3 nifi supergroup 3234 2026-05-10 15:23 /user/itversity/transactions/afde9c73-6dd5-4ca6-92c1-83c921d9efbb
-rw-r--r-- 3 nifi supergroup 1896 2026-05-10 15:21 /user/itversity/transactions/b85a50fb-b1dc-474e-bc90-89995702297f
-rw-r--r-- 3 nifi supergroup 2278 2026-05-10 15:21 /user/itversity/transactions/bc536274-fdb4-42ad-9bbc-fc29db0c5b11
-rw-r--r-- 3 nifi supergroup 2008 2026-05-10 15:21 /user/itversity/transactions/c70c6f5d-4b65-476e-0a75-0e745f168e6f
```

```
PS C:\Users\moham\Desktop\week 8> docker exec -it itvdelab_assignment bash
itversity@itvdelab:~$ hdfs dfs -ls /user/itversity/market_data
Found 23 items
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/065dece4-b189-4aa9-8cb2-375923bf04c3
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/073994f1-805c-4193-9b03-d2182f9e8a2d
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/0fb08479-b840-422f-9fc7-93d566a3390a
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:23 /user/itversity/market_data/1ad85150-d6a3-40d2-a749-14c106154772
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/26dae0c0-c7e7-44ca-8a3e-29a2de9e1e4e
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/29d15423-b74f-494b-a656-59c4cfff5b4
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:22 /user/itversity/market_data/305c112c-0280-4d18-87d8-e73e87102f1a
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/441b3a56-b041-4e07-8e8c-4aa5a0374a1
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/47a6cdcd-d02b-4ae4-8783-363cd82c6f4a
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/4c025ec7-7b06-4369-b752-bef92950d975
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/4c8041af-a183-44da-aac6-b232713309d1
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/55c44e26-b5d8-4b65-9020-b5fac213372b
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/6250c217-a4f5-410d-9cfc-36107a386d36
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/9ce7ac03-0a0b-405a-903f-9c555f74e223
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/a0c6e2df-2492-4a0e-8e77-58ec1a43d734
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/a121b0ed-988a-4018-ac31-93d8301bc91c
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/aa0dde3-91da-4390-82b8-7ac5b3bf9e49
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/d9d47256-46d7-47f1-b9e6-62d1eb059bec
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/e08cf518-f521-4d3b-938a-2b6453e633ba
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/e0b3f562-9758-49bf-872d-b5762693a215
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/f15e9cc0-25b8-46bc-8cdf-5b88ad517599
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/f5e10cb1-510e-44d1-8857-0d93d26f486e
-rw-r--r-- 3 nifi supergroup 435 2026-05-10 15:21 /user/itversity/market_data/fb77afcb-3edc-4dfa-9679-623103fd4e83
itversity@itvdelab:~$
```

12. Conclusion

This project provided practical experience in designing and implementing modern data engineering pipelines using Apache NiFi.

The assignment demonstrated how Apache NiFi can be used to build scalable and reliable real-time data pipelines using multiple data sources and distributed storage systems.

13. Technologies Used

- Apache NiFi 2.9.0
- Docker & Docker Compose
- PostgreSQL
- Hadoop HDFS
- Python
- JSON
- Apache NiFi Record Processors